

AJ C PIPATTANAKUN

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PERSONAL SUMMARY

Computer Science graduate with a Minor in Data Science, currently pursuing a master's degree in Computer Science. Experience working on academic and internship projects involving coding, debugging, backend services, API testing, software integration, and technical documentation. Comfortable using multiple high-level programming languages including C++, Java, Python, SQL, and basic C, with working knowledge of object-oriented programming and Object-Oriented Analysis and Design (OOAD).

Hands-on experience includes setting up local development environments, troubleshooting build and runtime issues, testing API health endpoints, validating JSON outputs, and checking database connectivity. Familiar with Windows and basic UNIX/Linux command-line environments, Git/GitHub version control, software testing concepts, regression testing practice, and requirement-based validation.

CORE SKILLS

Programming Languages: C++, Java, Python, SQL, basic C

Software Engineering: Coding, debugging, object-oriented programming, Object-Oriented Analysis and Design (OOAD), software design, and technical documentation

Software Testing: Unit testing concepts, regression testing, requirement testing, integration testing, API health checks, and validation of expected system behavior

Systems & Operating Systems: Windows 10/11, basic UNIX/Linux command line, environment setup, process checks, and software/hardware integration concepts

Version Control & Tools: Git, GitHub, Visual Studio Code, Docker, FastAPI, Node.js, Jupyter Notebooks; familiar with version control workflows similar to Bitbucket

Communication: Reading technical documentation, writing clear issue notes, explaining progress, and participating in team updates

EDUCATION

University of Massachusetts Lowell

Master of Science in Computer Science

Expected Spring 2028

Lowell, MA

- Relevant Courses: Advanced Computer Science, Software Development, Artificial Intelligence, Data Systems

University of Massachusetts Lowell

Bachelor of Science in Computer Science, Minor in Data Science

September 2020 - August 2024

Lowell, MA

- Relevant Courses: Machine Learning, Data Visualization, Statistical Modeling

WORK EXPERIENCE

Software Engineering Intern

Tihranix, Foxborough, MA

May 2026 – Present

- Worked on forecasting and decision intelligence features using FastAPI, Next.js, TypeScript, and Python.
- Improved the Forecast Builder so previews respond to selected KPIs, forecast models, time horizons, and sensitivity drivers.
- Connected forecast previews with PLDM-driven context and improved source labeling for live preview, static output, and fallback states.
- Added model- and KPI-specific preview logic to update forecast charts, reasoning, accuracy display, and active model state.
- Used Git, GitHub pull requests, Docker, VS Code, and local build validation as part of the team development workflow.

- Design and maintain a fully responsive website for the restaurant using HTML, CSS, and JavaScript, which enhanced the restaurant's online presence.
- Integrate online menu and contact forms, improving customer engagement and facilitating easier customer communication.
- Tools: HTML, CSS, JavaScript.

PROJECTS

Real-Time Multi-Sensor Monitoring and Fault-Tolerant System

January 2026

C++, Python, Linux, TCP/IP, CMake, Google Test

- Designing a multithreaded C++ application to process simulated real-time temperature, voltage, position, and motion sensor data over TCP/IP.
- Implementing object-oriented sensor interfaces, thread-safe message queues, fault detection, structured logging, and deterministic system-state transitions.
- Developing a Python-based sensor simulator and fault-injection tool to test timeouts, malformed messages, abnormal readings, and communication failures.
- Creating unit, integration, regression, and requirement-based tests using Google Test to verify expected system behavior.
- Configuring a Linux build workflow using CMake, GitHub Actions, and basic runtime diagnostic tools.

Reliable Distributed Command Processing System

April 2026

Java, Python, Linux, TCP/IP, JUnit, Gradle, Docker

- Designing a Java-based distributed system that exchanges structured commands between a control station and a simulated remote unit over TCP/IP.
- Applying object-oriented analysis and design principles through Command, State, Strategy, and Factory design patterns.
- Implementing message validation, acknowledgements, timeouts, retries, duplicate-message detection, sequence-number checks, and audit logging.
- Developing state-dependent command handling to reject invalid operations and maintain predictable system behavior.
- Creating JUnit unit, integration, regression, and requirement-based tests for message parsing, command execution, invalid state transitions, and network failures.

CERTIFICATIONS

Google IT Support Professional Certificate

Completed: January 2021

- Developed by Google and delivered through Coursera; consists of five courses focused on hands-on IT support training.
- Gained foundational skills in troubleshooting, customer service, networking, operating systems, system administration, and IT security.

UMass Global Data Science Bootcamp

Completed: August 2024

- Comprehensive data science training with a focus on business insights, structured thinking, predictive modeling, and advanced data visualization.
- Gained expertise in using issue trees, value driver trees, A / B testing, and advanced SQL for business analytics.